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JAVA PROJECT REPORT ON

Personal Health Record Monitoring System

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APRIL 2026

Semester: **II**

DECLARATION

I hereby declare that the project entitled "**Personal Health Record Monitoring System"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **Mr.Aravind** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

Sathya Rooba.M

CERTIFICATE

This is to certify that the project entitled **"Personal Health Record Monitoring System"** is a **bona fide work** carried out by **Sathya Rooba.M [BSC CS 2025-2026], KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:15/04/2026

Submitted for the Viva-Voce Examination held on __15/04/2026_____

Guide: **Head of the Department**(Technical Trainer, KGCAS) (Technical Head, KGCAS)

___Mr.Aravind_____
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ABSTRACT

The Personal Health Record Monitoring System is a Java-based application developed using Swing GUI to store and manage patient health details efficiently. In traditional healthcare environments, maintaining patient records manually is a difficult and time-consuming process that often leads to errors and data mismanagement. To overcome these issues, this system provides a digital solution for storing patient information in a structured and organized manner.

The system allows users to enter patient details such as name, age, disease, and weight through a user-friendly graphical interface. These details are stored using file handling techniques, which enables future retrieval of records whenever required. The application is designed using core Java programming concepts such as classes and objects to represent patient data, encapsulation to ensure data security, inheritance and polymorphism to enhance functionality, and exception handling to manage invalid inputs effectively.

Overall, this project demonstrates the practical implementation of Java programming concepts in developing a real-world application. It helps in reducing manual effort, improving accuracy, and providing an efficient way to manage patient health records digitally.

1. INTRODUCTION

In the modern healthcare system, maintaining patient records manually has become increasingly difficult due to the growing number of patients and the need for accurate data management. Manual record-keeping methods are time-consuming, prone to errors, and may lead to data loss or duplication. To address these challenges, a computerized system is required that can efficiently store, manage, and retrieve patient information.

The Personal Health Record Monitoring System is developed as a solution to these problems. It is a Java-based application that uses Swing GUI to provide an interactive interface for users. The system allows users to enter patient details such as name, age, disease, and weight, and stores this information in a file for future reference.

The application is developed using object-oriented programming concepts such as classes, objects, inheritance, encapsulation, and polymorphism. These concepts help in organizing the code efficiently and improving the flexibility of the system. Exception handling is also implemented to ensure that the system handles invalid inputs without crashing. Overall, this system improves efficiency, reduces manual effort, and provides a reliable way to manage patient records.

1.1 OBJECTIVE

The main objective of the Personal Health Record Monitoring System is to develop a simple, efficient, and user-friendly application that helps in storing and managing patient health records digitally. The system aims to reduce manual work and improve the accuracy of maintaining patient data. It also focuses on implementing core Java programming concepts such as classes, objects, inheritance, encapsulation, and polymorphism. Another objective is to provide a graphical user interface that allows users to easily interact with the system and manage records effectively. Additionally, the system aims to store data using file handling techniques so that patient records can be accessed in the future whenever required.

1.2 OVERVIEW

The Personal Health Record Monitoring System is a desktop-based application designed to simplify the process of managing patient information. It provides a graphical user interface using Java Swing, which allows users to enter details such as name, age, disease, and weight in a structured manner. Once the data is entered, the system processes and stores it in a file, making it available for future use.

The system also provides an option to view stored patient records. It is developed using object-oriented programming concepts, which help in maintaining a clear structure and improving code reusability. Exception handling is used to ensure that incorrect inputs are handled properly. File handling is used to store and retrieve patient data efficiently. Overall, the system provides a simple and effective solution for maintaining health records digitally.

1.3 FEATURES

The Personal Health Record Monitoring System provides several features that make it useful and efficient for managing patient data. The system includes a user-friendly graphical interface that allows users to easily enter patient details. It supports data storage using file handling, which ensures that records are saved and can be accessed later. The system also provides the ability to view stored records, making it easy to retrieve patient information when needed.

In addition, the system includes error handling mechanisms to manage invalid inputs and ensure smooth operation. It is designed to be simple and efficient, reducing manual work and minimizing errors. These features make the system suitable for small clinics and healthcare environments.

1. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. In this project, the need for a digital system to manage patient records was analyzed. The existing manual system has several drawbacks, such as difficulty in maintaining records,

chances of errors, and lack of proper data storage. This phase helps in designing a system that can overcome these problems and provide an efficient solution.

2.1 PROBLEM DEFINITION

In traditional healthcare systems, maintaining patient records manually is a difficult task that involves a lot of time and effort. Manual records are prone to errors, and retrieving patient information can be challenging. There is also a risk of losing important data. Therefore, there is a need for a computerized system that can store and manage patient data efficiently. This project aims to solve these problems by developing a digital health record monitoring system.

2.2 PROJECT OVERVIEW

The Personal Health Record Monitoring System is a Java-based application that allows users to store and manage patient information. The system provides a graphical interface where users can enter patient details such as name, age, disease, and weight. The entered data is stored using file handling techniques and can be retrieved whenever required. The system is designed using object-oriented programming concepts, which improve code structure and maintainability.

2.3 SCOPES AND LIMITATIONS

The scope of this system includes its use in small clinics and healthcare centers where maintaining digital records is necessary. It helps in storing patient data efficiently and retrieving it easily. However, the system has certain limitations, such as the absence of a database, single-user access, and lack of advanced features like online access or data security mechanisms. Despite these limitations, the system is useful for small-scale applications.

1. SYSTEM DESIGN

System design is an important phase in software development where the overall structure and working of the system are planned. In the Personal Health Record Monitoring System, the design focuses on creating a simple and user-friendly application that efficiently stores and manages patient data. The system is designed using Java Swing for the graphical user interface, which allows users to interact with the system easily.

The application follows an object-oriented approach, where patient details are represented using classes and objects. Each patient record contains attributes such as name, age, disease, and weight. The system ensures proper data handling using encapsulation, which protects the data from unauthorized access. Inheritance and polymorphism are used to improve code reusability and flexibility, making the system easy to extend in the future.

The system is divided into different modules to handle specific functionalities. The input module allows users to enter patient details through a graphical interface. The processing module handles the validation of input data and ensures that only correct information is stored. The storage module uses file handling techniques to save patient records permanently. The retrieval module allows users to view previously stored records whenever required.

The design also includes exception handling mechanisms to manage errors such as invalid inputs or file-related issues. This ensures that the system runs smoothly without crashing. Overall, the system design focuses on simplicity, efficiency, and reliability, making it suitable for small-scale healthcare applications.

3.1 MODULES

The Personal Health Record Monitoring System is divided into several modules, each responsible for performing a specific function. The input module is used to collect patient details such as name, age, disease, and weight from the user through a graphical user interface. The processing module validates the entered data and ensures that all required fields are filled correctly.

The storage module is responsible for saving patient records using file handling techniques, which allows data to be stored permanently. The retrieval module enables users to access and view stored patient records whenever needed. Additionally, the system includes an error handling module that manages exceptions and prevents the system from crashing due to invalid inputs or unexpected errors.

These modules work together to provide a complete system that efficiently manages patient health records.

1. METHODOLOGY

The methodology of the Personal Health Record Monitoring System describes the step-by-step process followed to design and develop the application. This project follows a systematic approach to ensure that the system is developed efficiently and meets all the required objectives. The methodology includes various stages such as problem analysis, system design, program development, testing, and documentation. Each stage plays a crucial role in building a reliable and user-friendly application.

Initially, the problem is analyzed to understand the difficulties faced in maintaining patient records manually. Based on this analysis, the system is designed to provide a digital solution using Java programming. The application is then developed using object-oriented programming concepts such as classes, objects, inheritance, polymorphism, and encapsulation. The graphical user interface is created using Java Swing to make the system easy to use.

After development, the system is tested with different inputs to ensure that it functions correctly and handles errors effectively. Exception handling techniques are used to manage invalid inputs and avoid system crashes. Finally, the results are documented, and the system is evaluated to ensure that it meets the objectives. This structured methodology helps in developing a clear, efficient, and reliable health record monitoring system.

4.1 PROBLEM ANALYSIS

Problem analysis is the first step in the methodology, where the existing system is studied to identify its limitations. In traditional methods, patient records are maintained manually, which is time-consuming and prone to errors. There is also a risk of losing important data, and retrieving records becomes difficult. This analysis helps in understanding the need for a computerized system that can store and manage patient data efficiently.

4.2 SYSTEM DESIGN

In this stage, the overall structure of the system is planned. The system is designed using Java Swing for the graphical user interface and object-oriented programming concepts for better organization of code. The system is divided into modules such as input, processing, storage, and retrieval. This design ensures that the system is easy to use and maintain.

4.3 PROGRAM DEVELOPMENT

Program development involves writing the actual code for the system. The application is developed using Java programming language, where classes and objects are used to represent patient data. Features such as file handling are implemented to store data, and exception handling is used to manage errors. The graphical user interface is created to allow users to interact with the system easily.

4.4 TESTING AND VALIDATION

Testing is performed to ensure that the system works correctly under different conditions. Various test cases are used to check whether the system accepts valid inputs and rejects invalid ones. The system is also tested for file handling operations to ensure that data is stored and retrieved correctly. Validation ensures that the system meets all the requirements.

4.5 RESULT AND DOCUMENTATION

In this stage, the results of the system are analyzed and documented. The system successfully stores and retrieves patient records, reducing manual work and improving efficiency. Proper documentation is prepared to explain the working of the system and to help users understand how to use it.

4.6 ALGORITHM

Step 1: Start the program

Step 2: Display the user interface

Step 3: Enter patient details (name, age, disease, weight)

Step 4: Validate the input data

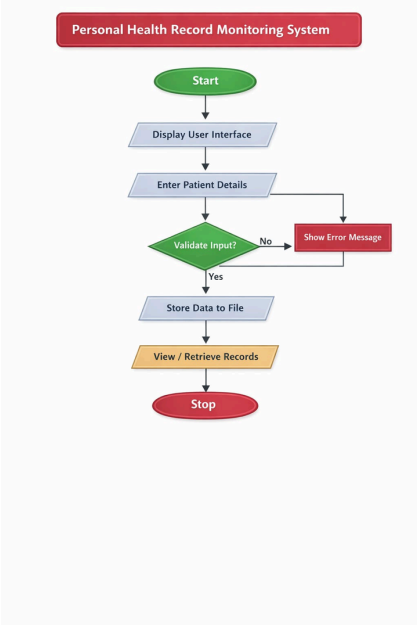
Step 5: Store the data using file handling

Step 6: Display confirmation message

Step 7: Retrieve data when required

Step 8: Stop the program

4.7 FLOW CHART



1. SOURCE CODE

```
// Package declaration package healthsystem;


import javax.swing.; import java.awt.; import java.awt.event.; import java.io.;


// Base class (Inheritance) class Person { protected String name; protected int age;


public Person(String name, int age) { // Constructor
    this.name = name;
    this.age = age;
}


public void displayInfo() { // Method
    System.out.println("Name: " + name + ", Age: " + age);
}


}


// Derived class (Inheritance + Encapsulation) class Patient extends Person { private String disease; private double weight;
```

```
public Patient(String name, int age, String disease, double weight) {
    super(name, age);
    this.disease = disease;
    this.weight = weight;
}

// Getters & Setters (Encapsulation)
public String getDisease() { return disease; }
public void setDisease(String disease) { this.disease = disease; }

public double getWeight() { return weight; }
public void setWeight(double weight) { this.weight = weight; }

// Method Overriding (Polymorphism)
@Override
public void displayInfo() {
    System.out.println("Patient: " + name + ", Age: " + age +
            ", Disease: " + disease + ", Weight: " + weight);
}

public String toFileString() {
    return name + "," + age + "," + disease + "," + weight;
}

}
```

```
// File Handling Class class FileManager { private static final String FILE_NAME = "patients.txt";

public static void savePatient(Patient p) throws IOException {
    FileWriter fw = new FileWriter(FILE_NAME, true);
    fw.write(p.toFileString() + "\n");
    fw.close();
}

public static String readPatients() throws IOException {
    BufferedReader br = new BufferedReader(new FileReader(FILE_NAME));
    StringBuilder data = new StringBuilder();
    String line;
    while ((line = br.readLine()) != null) {
        data.append(line).append("\n");
    }
    br.close();
    return data.toString();
}

}

// GUI Class (Swing) public class HealthGUI extends JFrame {

private JTextField nameField, ageField, diseaseField, weightField;
private JTextArea displayArea;

public HealthGUI() {
```

```
setTitle("Personal Health Record System");
setSize(400, 500);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new FlowLayout());

add(new JLabel("Name:"));
nameField = new JTextField(15);
add(nameField);

add(new JLabel("Age:"));
ageField = new JTextField(15);
add(ageField);

add(new JLabel("Disease:"));
diseaseField = new JTextField(15);
add(diseaseField);

add(new JLabel("Weight:"));
weightField = new JTextField(15);
add(weightField);

JButton saveBtn = new JButton("Save Record");
JButton viewBtn = new JButton("View Records");

add(saveBtn);
add(viewBtn);
```

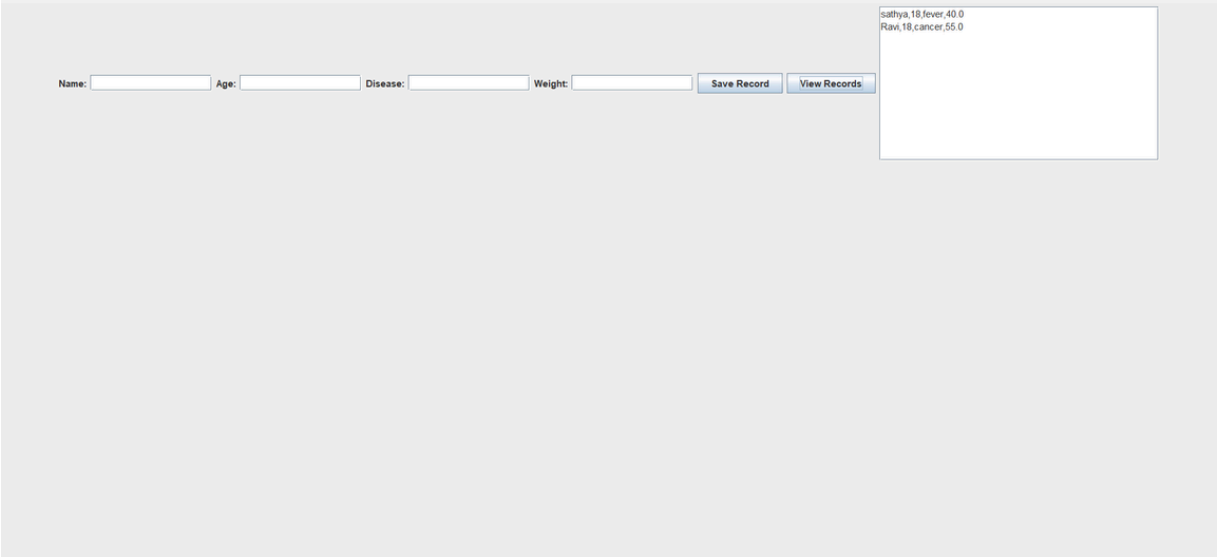
```
displayArea = new JTextArea(10, 30);

add(new JScrollPane(displayArea));


// Button Actions

saveBtn.addActionListener(new ActionListener
```

1. INPUT AND OUTPUT SCREENS



1. DISCUSSION & CONCLUSION

7.1 DISCUSSION

The Personal Health Record Monitoring System was developed to provide a simple and efficient way to store and manage patient data. The system allows users to enter patient details and store them digitally using file handling. During the development process, core Java concepts such as classes, objects, inheritance, polymorphism, and exception handling were implemented. The system was tested with different inputs to ensure that it works correctly and produces accurate results. The results show that the system reduces manual effort, minimizes errors, and improves efficiency in managing patient records.

7.2 CONCLUSION

The Personal Health Record Monitoring System successfully demonstrates the practical application of Java programming concepts in developing a real-world application. The system provides a simple and effective solution for storing and managing patient records digitally. It improves accuracy, reduces manual work, and enhances data management. The system can be further improved by adding features such as database integration, multi-user support, and advanced user interface design. Overall, this project helps in understanding how Java can be used to develop useful and efficient applications

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